Milestone 2 Report

Temporal Trend Visualization

Visualizing US Natural Disaster Declarations: Trends and Patterns

FEMA Web Disaster Declarations Dataset | Weeks 3 - 4

Project Phase	Focus Area	Duration
Milestone 2 of 4	Temporal Trend Analysis	Weeks 3 - 4 (2 weeks)

1. Milestone Overview

Milestone 2 builds on the cleaned FEMA Natural Disaster Declarations dataset produced in Milestone

1. The focus of this phase is temporal analysis — understanding how the frequency, variety, and nature of federally declared disasters have evolved across decades. By examining time-based trends, this milestone reveals critical patterns in disaster occurrence that are essential context for the geographical and incident-type analyses to follow.

2. Goals & Deliverables

2.1 Primary Goals

This milestone targets the following analytical objectives, drawn from the project specification:

- Develop a refined time-series plot showing total disaster declarations per year across the full dataset history.
- Create multi-line trend charts illustrating the change in major incidentType categories (flood, hurricane, fire, tornado, etc.) over time.
- Analyze seasonal patterns in disaster declarations — i.e., average declarations per month — to identify cyclical weather-driven events.
- Explore incident duration trends over time where date range data is available.
- Experiment with line charts, stacked area charts, and rolling-average overlays to best represent temporal dynamics.

3. Technical Approach

3.1 Data Aggregation

Using Pandas, the cleaned dataset from Milestone 1 is grouped and aggregated along time dimensions. Key operations include:

- Extracting year, month, and decade from the declarationDate column for temporal grouping.
- Grouping by year to obtain total declaration counts per year (declarations_per_year).
- Grouping by incidentType and year to enable per-type trend lines.
- Grouping by month across all years to identify seasonal patterns.
- Computing rolling 5-year averages using Pandas rolling() for smoothed trend visualization.

3.2 Visualization Libraries

The following Python libraries from the project tech stack are used in this milestone:

Library	Usage in Milestone 2
Pandas	*Data loading, groupby aggregation, resampling, and rolling averages*
Matplotlib	*Base rendering engine for all static time-series and bar charts*
Seaborn	*High-level styling for bar charts, line plots, and heatmaps of monthly data*
Plotly Express	*Interactive time-series charts with hover detail, zoom, and filtering*
NumPy	*Numerical operations and smoothing calculations*

4. Expected Temporal Visualizations

The four core visualizations produced in this milestone each address a distinct facet of temporal disaster behavior. Together they form a comprehensive temporal portrait of FEMA-declared disasters across US history.

4.1 Total Disaster Declarations Per Year

A refined line chart plotting annual declaration counts from the earliest year in the dataset to the most recent. A rolling 10-year average overlay smooths short-term fluctuations and surfaces the long-term trend.

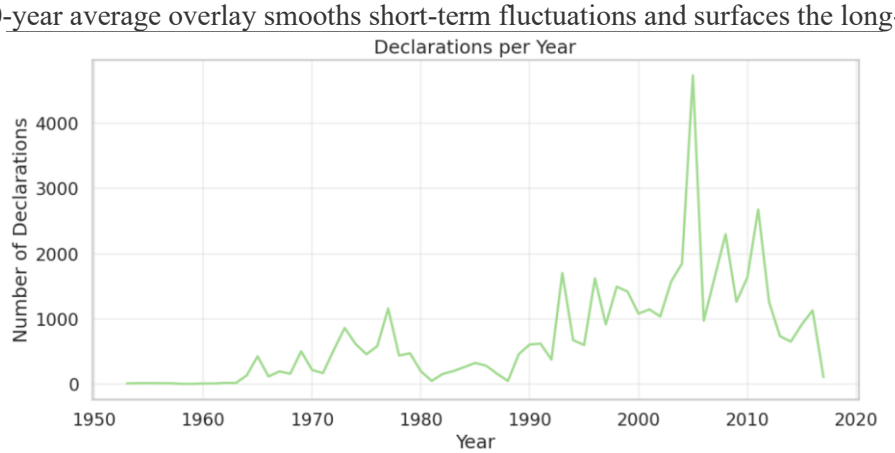

Figure 1. *Time-series of annual FEMA disaster declarations with 5-year rolling average overlay. Upward trends post-1980 are expected as federal disaster policy expanded and data quality improved.*

4.2 Incident Type Trends Over Time

A chart where each line represents a major incidentType category (e.g., Flood, Hurricane, Fire, Tornado, Severe Storm). This visual reveals which disaster types are increasing, stable, or declining over time, and allows comparison of their relative frequencies.

- Line colors are mapped to distinct incident types for clarity.
- An interactive Plotly version supports hover detail showing exact counts per year.
- Top 6–8 incident types by overall frequency are selected to avoid over-crowding.

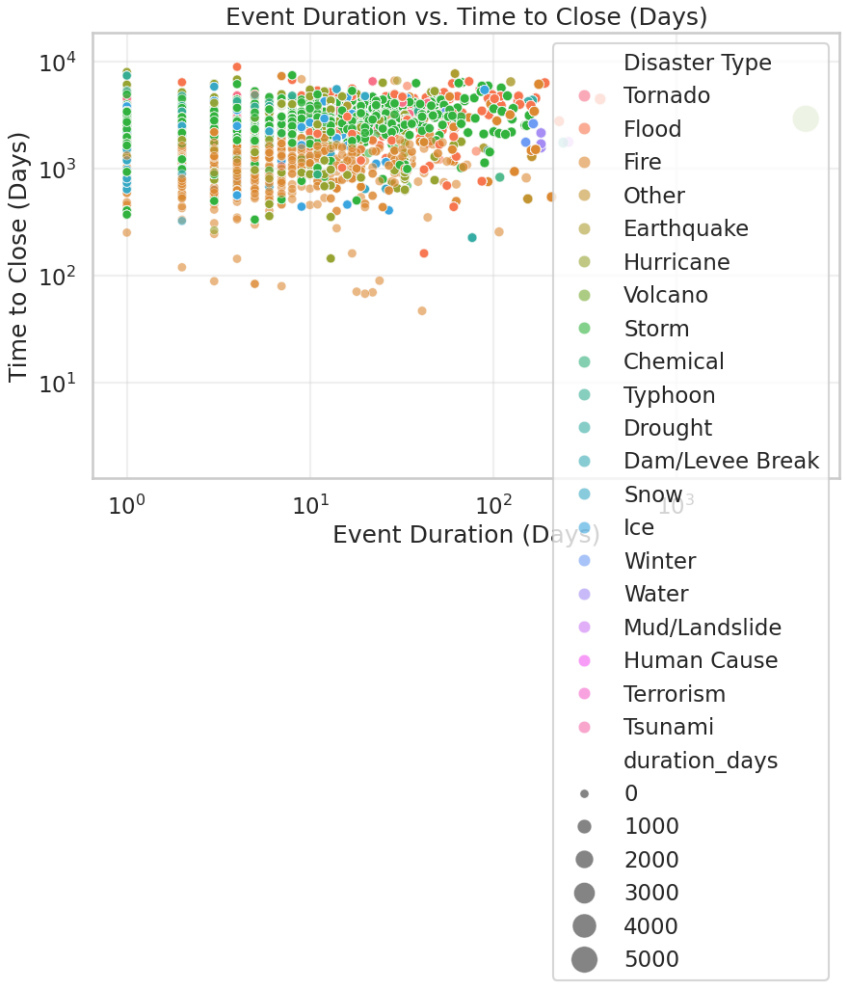

Figure 2. *Multi-line trend chart segmented by incidentType. Floods and Severe Storms are anticipated to dominate, with Hurricane activity concentrated in distinct temporal clusters.*

4.3 Seasonal Pattern Analysis

Using Seaborn, a bar chart shows the average number of disaster declarations per calendar month, aggregated across all years. This exposes the seasonality of disaster events — for example, hurricane season peaking in late summer and early fall.

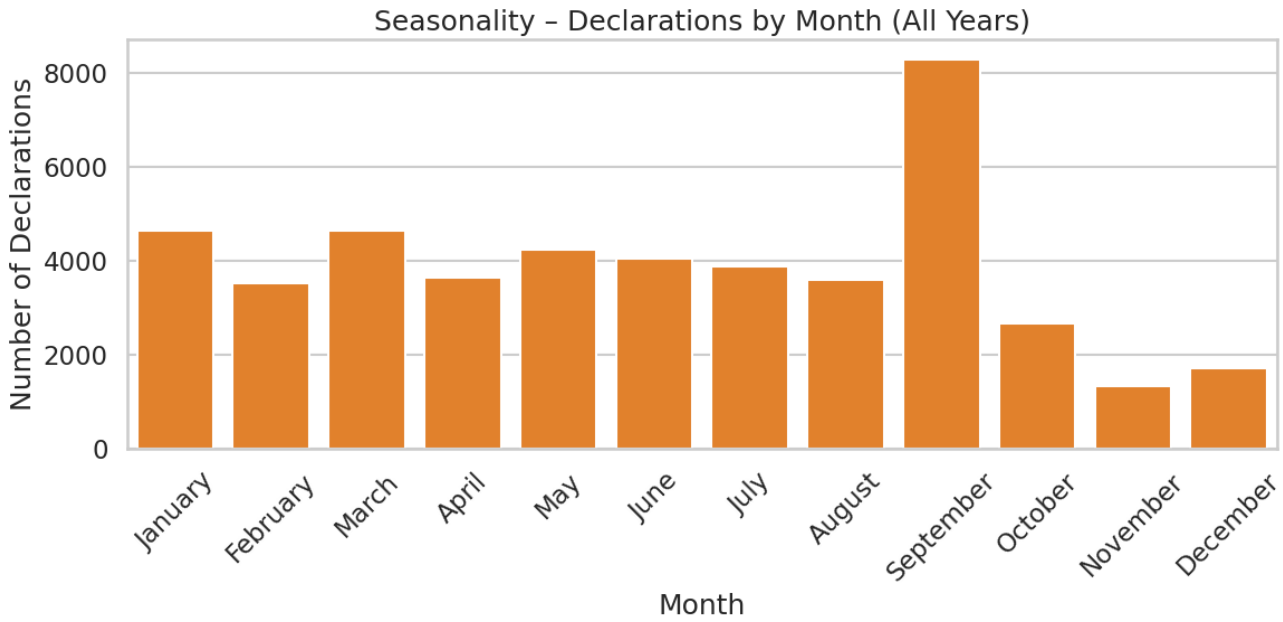

Figure 3. *Monthly seasonality chart revealing cyclical disaster patterns. Spring flooding (March–May) and Atlantic hurricane season (September) are expected to produce visible peaks.*

4.4 Stacked Area Chart (Share of Disaster Types Over Time)

This chart analyzes how the proportion (share) of different disaster types has changed across years in the United States. Instead of showing raw counts, the chart shows what percentage each disaster type contributed in each year, helping us understand:

- Which disaster types dominated in different time periods
- How the composition of disasters shifted over decades
- Long-term rising/falling trends of specific hazards
- Relative importance of droughts, floods, storms, hurricanes, tornadoes, snow, ice, and fire

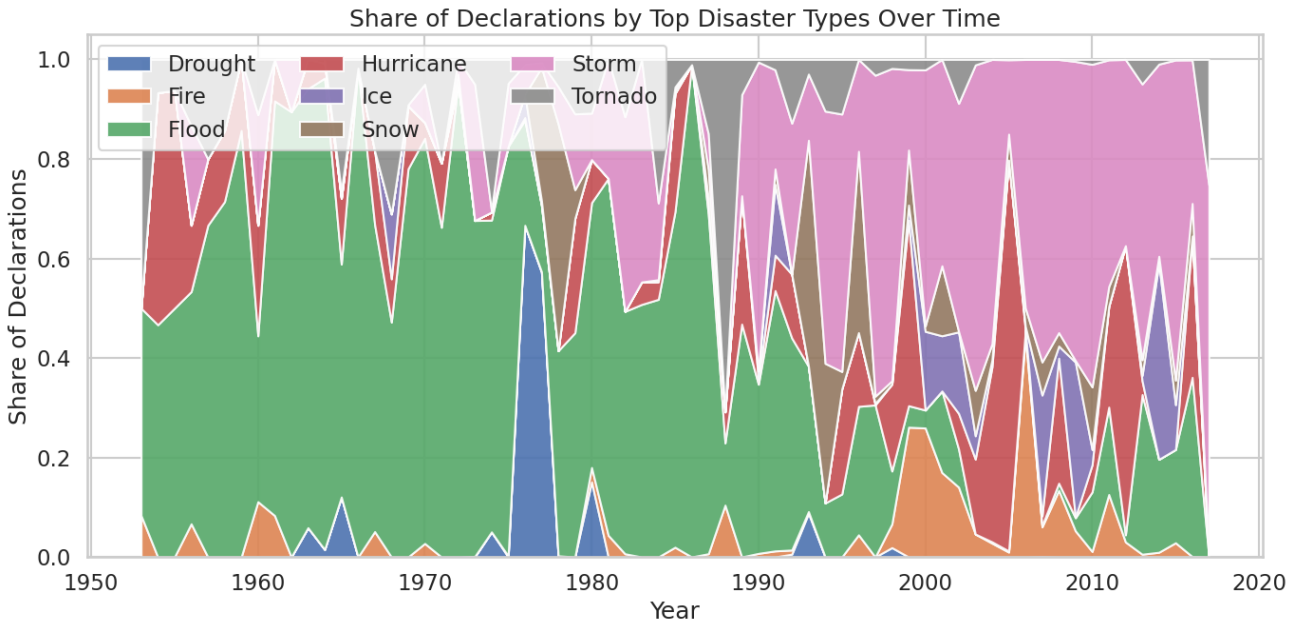

Figure 4. *The chart shows how the share of different disaster types has shifted over the past 70 years, with floods and storms dominating most periods while other hazards appear only occasionally, revealing long-term changes in U.S. natural disaster patterns.*

5. Evaluation Plan

Per the project milestone plan, the following evaluation criteria are applied at the end of Week 4 to assess successful completion of Milestone 2:

Metric	*Target*	*Evaluation Method*
Visualization Clarity	*All charts readable at standard resolution with no overlapping labels*	*Peer review + visual inspection*
Trend Representation Accuracy	*Counts match source data when cross-referenced with Pandas aggregations*	*Spot-check 5 year-points against raw data*
Chart Type Appropriateness	*Each visualization type matches its data shape and analytical purpose*	*Reviewer assessment against best practices*
Label Completeness	*All axes, legends, titles, and data sources are labeled on every chart*	*Checklist: title, x-axis, y-axis, legend, source*
Deliverable Count	*Minimum 3–4 distinct temporal visualizations produced*	*Count of final outputs in notebook*

5.1 Success Criteria Summary

- At least 3–4 distinct temporal visualizations are finalized and included in the Jupyter Notebook.
- All visualizations are well-labeled with accurate axis titles, clear legends, and chart titles.
- Trends — upward, cyclical, or stable — are clearly visible and directly interpretable.
- Chart types are appropriate for the data: line/area for time series, bar/heatmap for categorical-temporal distributions.
- One or more visualizations include an interactive Plotly version for dynamic exploration.

6. Conclusion

The temporal trend analysis reveals clear shifts in U.S. natural disaster patterns over time, showing rising variability, increasing dominance of storms and floods, and changing seasonal and long-term behaviors. These visual insights form the foundation for deeper geographic and incident-type analysis in the next milestone.